

Modern Hamiltonian Monte Carlo for Structural State-Space Inference

A chapter from the BayesFilter monograph

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Chapter 1

Modern Hamiltonian Monte Carlo for Structural State-Space Inference

A posterior distribution may take a long time to explore because its geometry is difficult, because each likelihood evaluation is expensive, or because the sampler uses its evaluations poorly. These explanations lead to different repairs. A nonlinear change of coordinates can remove a funnel without making a Kalman recursion cheaper. A parallel filter can reduce evaluation latency without making a poorly chosen trajectory explore the posterior. A new trajectory rule can reduce gradient work without correcting an inaccurate likelihood. For structural state-space inference, the useful question is therefore which combination of geometry, dynamics, adaptation and filtering produces accurate estimates of economically meaningful quantities at a reasonable total cost.

Recent work offers several answers. ChEES and SNAPER learn trajectory lengths that suit parallel computation; MALT and MEADS use persistent or partially refreshed dynamics; MAMS changes the dynamics to a microcanonical construction; and LAPS uses an unadjusted ensemble transient before switching to adjusted sampling. Neural transport and Pathfinder address geometry and initialization. Parallel Gaussian message passing and gradient-informed particle methods exploit the state-space structure itself [Hoffman et al., 2021, Sountsov and Hoffman, 2022, Riou-Durand and Vogrinc, 2023, Hoffman and Sountsov, 2022, Robnik et al., 2025, Robnik and Seljak, 2026, Hoffman et al., 2019, Zhang et al., 2022, Särkkä and García-Fernández, 2020, Corenflos and Finke, 2024].

This chapter develops the mathematical constructions needed to assess these methods. The central arguments are given in full: the probability measure being sampled, the integration and acceptance corrections, the quantities adapted, and the diagnostic assumptions. General ergodicity and optimal-scaling theorems are cited with their conditions rather than asserted for arbitrary economic models. The discussion covers work published or revised during approximately 2016–2026, with older HMC and particle-MCMC foundations included where they are needed to make the account self-contained. Author benchmarks are evidence about their stated experiments. They are not measurements of BayesFilter, MacroFinance or `dsge_hmc`.

1.1 The posterior, its coordinates and its computational cost

1.1.1 Marginal parameters and latent trajectories are different targets

Let $y_{1:T}$ be observations, $s_{0:T}$ latent states and ϑ structural parameters. An initial density m_0 , transition density m_t and observation density g_t define

$$p(\vartheta, s_{0:T} \mid y_{1:T}) \propto p(\vartheta) m_0(s_0 \mid \vartheta) \prod_{t=1}^T m_t(s_t \mid s_{t-1}, \vartheta) g_t(y_t \mid s_t, \vartheta), \quad (1.1)$$

$$L(\vartheta) = \int m_0(s_0 \mid \vartheta) \prod_{t=1}^T m_t(s_t \mid s_{t-1}, \vartheta) g_t(y_t \mid s_t, \vartheta) ds_{0:T}, \quad (1.2)$$

$$p(\vartheta \mid y_{1:T}) \propto p(\vartheta) L(\vartheta). \quad (1.3)$$

Sampling (1.1) introduces an unknown at every state and time. Sampling (1.3) can be low-dimensional even when the state vector and observation horizon are large. For a linear Gaussian model, Kalman filtering evaluates the integral exactly in real arithmetic. A nonlinear filter instead defines an approximation $\hat{L}(\vartheta)$. An exact MCMC transition for $p(\vartheta)\hat{L}(\vartheta)$ does not turn that approximation into $p(\vartheta)L(\vartheta)$. Sampler error and likelihood approximation error must be analyzed separately.

The prediction-error decomposition makes the cost visible:

$$\ell(\vartheta) = \log L(\vartheta) = \sum_{t=1}^T \log p(y_t \mid y_{1:t-1}, \vartheta). \quad (1.4)$$

An integrator may evaluate $\nabla \ell$ hundreds of times for one retained transition. Reducing the time per evaluation and reducing the number of evaluations are distinct opportunities. A useful accounting is

$$C_{\text{total}} = C_{\text{compile}} + C_{\text{initialize/map}} + C_{\text{adapt/verify}} + C_{\text{retained}}. \quad (1.5)$$

Each term includes likelihood, derivative, integration, diagnostic and host work. When B chains are batched, write $c_{\nabla}(B)$ for the measured latency of one batched gradient call. Neither $c_{\nabla}(B) = c_{\nabla}(1)$ nor $c_{\nabla}(B) = Bc_{\nabla}(1)$ should be assumed. Accelerator advantages depend on the tensor shapes, arithmetic intensity, memory and control flow of the actual target [Sountsov et al., 2024].

1.1.2 Constraints and a frozen transport

Suppose the model chart $\vartheta = C(q)$ maps $q \in \mathbb{R}^d$ bijectively to the admissible parameter space. A further frozen transport $q = F(z)$ may improve geometry. Assume the maps are differentiable with nonsingular Jacobians on the domain of use. Both maps act between d -dimensional coordinate domains, so $J_C, J_F \in \mathbb{R}^{d \times d}$. Assume twice differentiability when differentiating their log determinants. With J_C and J_F denoting those Jacobians, change of variables gives

$$\log \tilde{\pi}_q(q) = \log p(C(q)) + \ell(C(q)) + \log |\det J_C(q)|, \quad (1.6)$$

$$\log \tilde{\pi}_z(z) = \log \tilde{\pi}_q(F(z)) + \log |\det J_F(z)|. \quad (1.7)$$

The tilde indicates that a normalization constant is unnecessary. The score must differentiate the same expression. The chain rule and Jacobi's determinant formula imply

$$\nabla_z \log \tilde{\pi}_z(z) = J_F(z)^\top \nabla_q \log \tilde{\pi}_q(F(z)) + \nabla_z \log |\det J_F(z)|, \quad (1.8)$$

$$\partial_{z_j} \log |\det J_F(z)| = \text{tr}\{J_F(z)^{-1} \partial_{z_j} J_F(z)\}. \quad (1.9)$$

To obtain the second line, use $d \det J = \det J \text{tr}(J^{-1} dJ)$ and divide by $\det J$; its sign is locally constant when J is nonsingular. Freezing the learned coefficients of F removes adaptation during sampling. It does not remove the derivative through $F(z)$ or the log determinant.

For example, the positive chart $\vartheta = e^q$ adds q to the log density. Its score is $e^q \partial_\vartheta [\log p(\vartheta) + \ell(\vartheta)] + 1$. Omitting the 1 computes the score of a different density. A numerical acceptance statistic cannot establish that the intended target was used.

1.2 Corrected HMC and the scope of its guarantee

1.2.1 Hamiltonian invariance

From this point let $x \in \mathbb{R}^d$ denote the coordinates actually sampled, let $\pi(x) \propto e^{-U(x)}$ be normalizable, and suppose U is twice continuously differentiable wherever the smooth-flow argument is used. Assume the exact flow exists over the time interval under consideration. Introduce $p \in \mathbb{R}^d$ with $p \sim \mathcal{N}(0, M)$ and fixed symmetric positive-definite mass $M \in \mathbb{R}^{d \times d}$. The joint density is proportional to $e^{-H(x,p)}$, where

$$H(x, p) = U(x) + K(p), \quad K(p) = \frac{1}{2} p^\top M^{-1} p. \quad (1.10)$$

Hamilton's equations are

$$\dot{x} = M^{-1} p, \quad \dot{p} = -\nabla U(x). \quad (1.11)$$

Along a solution,

$$\frac{dH}{dt} = \nabla U(x)^\top M^{-1} p - (M^{-1} p)^\top \nabla U(x) = 0. \quad (1.12)$$

The phase-space velocity has divergence zero: $M^{-1} p$ has no x dependence and $-\nabla U(x)$ has no p dependence. Liouville's formula then gives a unit Jacobian for the flow. Energy and volume preservation together preserve $e^{-H} dx dp$. Momentum reversal $R(x, p) = (x, -p)$ reverses time because K is even. These are the foundations of HMC [Neal, 2011, Sections 2 and 3].

The leapfrog approximation consists of three shears:

$$p_{1/2} = p - \frac{\epsilon}{2} \nabla U(x), \quad (1.13)$$

$$x' = x + \epsilon M^{-1} p_{1/2}, \quad (1.14)$$

$$p' = p_{1/2} - \frac{\epsilon}{2} \nabla U(x'). \quad (1.15)$$

The kick Jacobian is $\begin{pmatrix} I & 0 \\ -\epsilon \nabla^2 U/2 & I \end{pmatrix}$ and the drift Jacobian is $\begin{pmatrix} I & \epsilon M^{-1} \\ 0 & I \end{pmatrix}$. Both determinants equal one. The palindromic ordering also gives $R \Phi_\epsilon R = \Phi_\epsilon^{-1}$. Consequently $\Psi = R \Phi_\epsilon^L$ is an involution. Numerical energy error remains; the next argument explains how to correct it.

1.2.2 Metropolis correction for a general involution

Proposition 1.1 (Deterministic Metropolis correction). *Let $\rho(w)$ be a density relative to a specified volume measure, and let Ψ be a differentiable bijection with $\Psi^2 = I$. Write $J(w) = |\det D\Psi(w)|$ in local coordinates for that measure. The proposal $w' = \Psi(w)$, accepted with*

$$a(w) = \min \left\{ 1, \frac{\rho(\Psi(w))J(w)}{\rho(w)} \right\}, \quad (1.16)$$

and otherwise left at w , satisfies detailed balance.

Proof. Differentiating $\Psi(\Psi(w)) = w$ gives $J(\Psi(w))J(w) = 1$. Hence

$$\begin{aligned} \rho(w)a(w) &= \min\{\rho(w), \rho(\Psi(w))J(w)\} \\ &= \rho(\Psi(w))a(\Psi(w))J(w). \end{aligned}$$

Multiplication by dw , followed by the change of variables $w' = \Psi(w)$, equates accepted probability flow in both directions. Rejected probability stays at the same state and is automatically symmetric. $\square$

For leapfrog, $J = 1$ and $\rho \propto e^{-H}$, so

$$a = \min\{1, e^{-\Delta H}\}, \quad \Delta H = H(x', p') - H(x, p). \quad (1.17)$$

The proof shows why a different integrator with non-unit Jacobian needs a different correction. It also shows the limit of the guarantee: invariance means that a stationary input remains stationary. It does not state how quickly arbitrary initial states approach stationarity or whether a finite run visits every relevant mode.

1.2.3 A deterministic force need not be a gradient

Replace $-\nabla U(x)$ in a kick by a deterministic position-only field $b(x)$. The map $(x, p) \mapsto (x, p + \hbar b(x))$ still has triangular Jacobian with determinant one, inverse given by $-\hbar$, and the required momentum-reversal symmetry. Its symmetric composition with a drift can therefore be corrected against the actual endpoint Hamiltonian using (1.17). It need not be a symplectic Hamiltonian integrator, and it need not be efficient, but the volume-and-involution proof does not require b to be a gradient.

This observation does not authorize an arbitrary force in every HMC extension. Energy conservation in (1.12), the microcanonical stationary-density calculation below, and score-based equipartition identities use the actual gradient of the declared potential. History-dependent, noisy or statefully changing forces also need an enlarged state and a new reversibility argument. The inspected BGS documentation in `dsge_hmc` declares an endpoint UKF target and a position-only field not asserted to be its exact score. Its mechanics must be assessed under that contract rather than silently reclassified as an exact-gradient method.

1.3 Trajectory length, resonance and learned adaptation

1.3.1 A Gaussian calculation that separates acceptance from exploration

Take $U(x) = x^2/(2\sigma^2)$ and unit mass. Solving the harmonic oscillator in (1.11) gives

$$x_\tau = x_0 \cos(\tau/\sigma) + \sigma p_0 \sin(\tau/\sigma), \quad (1.18)$$

$$p_\tau = p_0 \cos(\tau/\sigma) - \frac{x_0}{\sigma} \sin(\tau/\sigma). \quad (1.19)$$

With fresh $p_0 \sim \mathcal{N}(0, 1)$ at each transition, the stationary position chain is an AR(1) process with coefficient $r = \cos(\tau/\sigma)$. For $|r| < 1$,

$$\text{Corr}(X_0, X_k) = r^k, \quad \text{Corr}(X_0^2, X_k^2) = r^{2k}. \quad (1.20)$$

To obtain the second identity, a centered bivariate Gaussian pair with variance σ^2 and correlation r^k satisfies $\mathbb{E}[X_0^2 X_k^2] = \sigma^4(1 + 2r^{2k})$; subtract σ^4 and divide by $\text{Var}(X^2) = 2\sigma^4$. Summing the geometric series gives integrated autocorrelation times

$$\tau_{\text{int}}(X) = \frac{1+r}{1-r}, \quad \tau_{\text{int}}(X^2) = \frac{1+r^2}{1-r^2}. \quad (1.21)$$

Near a half-period, $r \approx -1$: the mean can look unusually efficient while the squared quantity mixes very slowly. At an exact half-period $X' = -X$, and at a full period $X' = X$. These limiting chains are not covered by the summable-correlation formulas. Both have acceptance one under exact integration, yet neither explores the distribution of $|X|$.

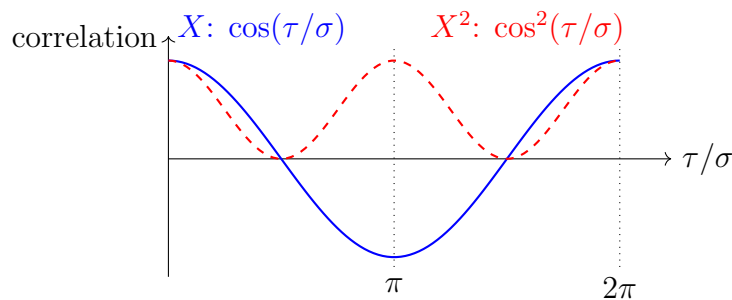


Figure 1.1: Exact Gaussian dynamics. The curves are analytic lag-one correlations, not benchmark results. Half-period trajectories change the sign of a draw but preserve its square; full-period trajectories preserve both.

1.3.2 Random lengths and NUTS

If P_τ preserves π for each fixed τ , and G is a state-independent distribution over lengths, then

$$\int \pi(dx) \int P_\tau(x, A) G(d\tau) = \int \pi(A) G(d\tau) = \pi(A). \quad (1.22)$$

Thus randomizing length preserves invariance without a new correction, provided each component kernel is valid and the distribution is fixed during retained sampling. It spreads the risk of resonance. If $\tau \sim \text{Unif}(0, 2\bar{\tau})$ independently at every Gaussian transition, direct integration yields

$$r_1 = \mathbb{E}[\cos(\tau/\sigma)] = \frac{\sin(2a)}{2a}, \quad (1.23)$$

$$r_2 = \mathbb{E}[\cos^2(\tau/\sigma)] = \frac{1}{2} + \frac{\sin(4a)}{8a}, \quad a = \bar{\tau}/\sigma. \quad (1.24)$$

Here X and $X^2 - \sigma^2$ are eigenfunctions of the mixture with eigenvalues r_1 and r_2 . In particular, choosing $2\bar{\tau} = 2\pi\sigma$ gives $r_1 = 0$ but $r_2 = 1/2$: the mean is decorrelated in one step while variance estimation still has integrated autocorrelation time three.

NUTS uses the observed trajectory to decide how far to simulate [Hoffman and Gelman, 2014, Section 3.1 and Algorithm 3]. In Euclidean coordinates the familiar turning conditions involve $(x_+ - x_-)^\top M^{-1}p_-$ and $(x_+ - x_-)^\top M^{-1}p_+$. Simply stopping when one becomes negative and accepting the endpoint is not the NUTS algorithm: a state-dependent stopping rule is not covered by (1.22). Tree building, candidate selection and the exclusion of asymmetric subtrees provide the required balance. That machinery removes a difficult manual length choice but produces irregular work across chains. The extent of the resulting accelerator cost depends on the implementation and target; it is not determined by the name NUTS alone.

1.3.3 ChEES and the stationary squared-jump identity

For a square-integrable function f , define

$$D_f = \frac{1}{2}\mathbb{E}[(f(X') - f(X))^2]. \quad (1.25)$$

At stationarity, $f(X)$ and $f(X')$ have equal means and variances, so expansion of the square gives

$$D_f = \text{Var}_\pi(f) - \text{Cov}(f(X), f(X')) = \text{Var}_\pi(f)\{1 - \rho_f(1)\}. \quad (1.26)$$

This connects a local movement objective to one-step correlation. It does not determine every higher-lag correlation. In a reversible stationary chain, a spectral representation makes the limitation precise. Normalize the spectral measure μ_f to have total mass one. If the relevant sums are integrable, then

$$\rho_f(1) = \int \lambda \mu_f(d\lambda), \quad \tau_{\text{int}}(f) = \int \frac{1+\lambda}{1-\lambda} \mu_f(d\lambda). \quad (1.27)$$

The function $(1+\lambda)/(1-\lambda)$ is convex on $(-1, 1)$ because its second derivative is $4/(1-\lambda)^3$. Jensen's inequality therefore implies

$$\text{ESS}_f = \frac{n}{\tau_{\text{int}}(f)} \leq n \frac{1 - \rho_f(1)}{1 + \rho_f(1)}. \quad (1.28)$$

The right side is an upper bound, not a guarantee that maximizing a jump criterion maximizes actual ESS [Sountsov and Hoffman, 2022, Appendix B].

ChEES uses the centered squared radius

$$f_{\text{ChEES}}(x) = \frac{1}{2}\|x - \mu\|^2, \quad \mu = \mathbb{E}_\pi[X]. \quad (1.29)$$

With the convention in (1.25), the original paper’s ChEES criterion is $2D_{f_{\text{ChEES}}}$; this constant factor does not change its maximizer. The center is estimated during adaptation. For independent Gaussian coordinates with variances σ_i^2 , the calculation above gives

$$D_{f_{\text{ChEES}}}(\tau) = \frac{1}{2} \sum_i \sigma_i^4 \sin^2(\tau/\sigma_i). \quad (1.30)$$

To derive the coefficient, the variance of $X_i^2/2$ is $\sigma_i^4/2$, its lag-one correlation is $\cos^2(\tau/\sigma_i)$, and cross-coordinate terms vanish. Larger variance directions receive weight σ_i^4 , but sufficiently many moderate directions can still outweigh one slow direction. Also, a large jump obtained using a long trajectory may be poor value per gradient. These motivate SNAPER [Hoffman et al., 2021, Sountsov and Hoffman, 2022].

1.3.4 SNAPER, endpoint derivatives and freezing

Let w be a unit principal direction estimated from the current ensemble. SNAPER uses

$$f_w(x) = \{w^\top(x - \mu)\}^2, \quad C_w(\tau) = \frac{D_{f_w}(\tau)}{\tau}. \quad (1.31)$$

Dividing by τ accounts for integration cost at fixed epsilon; optimizing along w focuses on a difficult direction rather than the sum of all directions. Both changes remain proxies. The principal covariance direction need not be the most difficult nonlinear or multimodal direction.

The pathwise derivative can be obtained without differentiating every integration step. For an exact trajectory with initial state held fixed, $dx_\tau/d\tau = M^{-1}p_\tau$. Holding w and μ fixed while differentiating,

$$\begin{aligned} \nabla f_w(x) &= 2w w^\top(x - \mu), \\ \frac{d}{d\tau} \frac{\{f_w(x_\tau) - f_w(x_0)\}^2}{2\tau} &= \frac{f_w(x_\tau) - f_w(x_0)}{\tau} \nabla f_w(x_\tau)^\top M^{-1}p_\tau - \frac{\{f_w(x_\tau) - f_w(x_0)\}^2}{2\tau^2}. \end{aligned} \quad (1.32)$$

$$(1.33)$$

This is the quotient rule plus the endpoint velocity. It explains why a trajectory objective need not require filter Hessians. A numerical algorithm uses an annotated endpoint derivative and treats the rounded step count and accept/reject behavior carefully; the exact derivative above is not an exact derivative of a discontinuous rounded-and-rejected program [Sountsov and Hoffman, 2022, Appendix C]. Such gradients propose adaptation updates. They do not enter the acceptance ratio.

A principal direction can be estimated by stochastic power iteration. If $z = x - \mu$, then $\mathbb{E}[zz^\top w] = \text{Cov}(X)w$. An Oja-type update takes the tangent component

$$\tilde{w} = w + \eta\{zz^\top w - w(w^\top z)^2\}, \quad w^+ = \tilde{w}/\|\tilde{w}\|. \quad (1.34)$$

Its expectation moves toward covariance eigenvectors, subject to the usual eigenvalue separation and stochastic-approximation conditions. An estimated center, diagonal mass, length parameters and epsilon all change the transition during adaptation. Their final

values must be frozen before the retained phase is interpreted using a fixed-kernel argument.

TensorFlow Probability 0.25.0 provides the kernel `SNAPERHamiltonianMonteCarlo`. It adapts diagonal scaling, the principal direction and trajectory length, but not epsilon; the `sample_snaper_hmc` helper supplies dual averaging. The paper’s SGA implementation uses its own synthetic acceptance update, so the helper should not be described as byte-for-byte reproduction of that tuning scheme. The implementation requires at least two chains. Its concentrated-start recommendation is an adaptation choice, not a justification for discarding independent dispersed verification starts. Source availability reduces porting effort; complete target, graph, GPU and checkpoint compatibility still needs testing.

1.4 Partial refreshment, MALT and MEADS

1.4.1 The Ornstein–Uhlenbeck refreshment

An underdamped Langevin process with mass M and scalar friction γ can be written

$$dX_t = M^{-1}P_t dt, \quad dP_t = -\nabla U(X_t) dt - \gamma P_t dt + \sqrt{2\gamma}M^{1/2}dW_t. \quad (1.35)$$

During a refreshment substep with X fixed, the linear stochastic equation has solution

$$P' = aP + \sqrt{1-a^2}M^{1/2}\xi, \quad a = e^{-\gamma h}, \quad \xi \sim \mathcal{N}(0, I). \quad (1.36)$$

Indeed, multiply the momentum equation by $e^{\gamma t}$ and integrate. The stochastic integral has covariance $2\gamma M \int_0^h e^{-2\gamma(h-s)}ds = (1 - e^{-2\gamma h})M$. If $P \sim \mathcal{N}(0, M)$, then $\text{Var}(P') = a^2M + (1-a^2)M = M$. Moreover (P, P') is jointly Gaussian with equal marginal covariances and symmetric cross-covariance aM , so its law is unchanged by exchange. The refreshment kernel Q therefore satisfies

$$e^{-K(p)}Q(p, dp') = e^{-K(p')}Q(p', dp). \quad (1.37)$$

The parameter a trades momentum persistence against randomization. It does not remove the need to choose a numerical step size.

1.4.2 Why MALT sums deterministic energy errors

MALT composes refreshment and leapfrog substeps in a symmetric Langevin trajectory [Riou-Durand and Vogrinc, 2023, Sections 2–3]. Consider the complete realized path. Its probability ratio under time reversal contains the endpoint density ratio $e^{-[H_{\text{end}} - H_{\text{start}}]}$, the Jacobians of deterministic maps, and the ratios of refreshment probabilities. The deterministic leapfrog Jacobians are one. From (1.37), each refreshment contributes

$$\frac{Q(p', dp)}{Q(p, dp')} = e^{K(p') - K(p)}. \quad (1.38)$$

Decompose the total energy change into changes during deterministic leapfrog substeps, ΔH_j^D , and refreshment changes ΔK_j^O . The path log ratio becomes

$$-\Delta H_{\text{total}} + \sum_j \Delta K_j^O = -\sum_j \Delta H_j^D. \quad (1.39)$$

The OU terms cancel rather than being counted as numerical energy errors. A reversal of the complete path, with the momentum sign convention and palindromic refreshment schedule, provides the extended-space involution. Accepting with $1 \wedge \exp(-\sum_j \Delta H_j^D)$ is therefore the relevant correction. Full momentum refresh at trajectory boundaries allows the accepted position transition to be used without inheriting a persistent rejected momentum.

For unit mass, the deterministic local error can be expressed using positions alone. Let $y = x + hv_{1/2}$, and write $g_x = \nabla U(x)$, $g_y = \nabla U(y)$. The endpoint velocities of the leapfrog substep are

$$v_0 = (y - x)/h + (h/2)g_x, \quad v_1 = (y - x)/h - (h/2)g_y. \quad (1.40)$$

Expanding the squared norms gives

$$E_h(x, y) = U(y) - U(x) + \frac{1}{2}(\|v_1\|^2 - \|v_0\|^2) \quad (1.41)$$

$$= U(y) - U(x) - \frac{1}{2}(y - x)^\top (g_y + g_x) + \frac{h^2}{8}(\|g_y\|^2 - \|g_x\|^2). \quad (1.42)$$

This reproduces the paper’s local error, equation (5). Summing it along the path yields the acceptance exponent. Using only the total endpoint Hamiltonian difference, including refreshment randomness, computes a different ratio.

Adaptive MALT estimates scaling, damping, epsilon and trajectory length [Riou-Durand et al., 2023]. Its empirical results distinguish efficiency per iteration from efficiency per gradient. It is sometimes slightly less gradient-efficient than uniformly randomized HMC using SNAPER adaptation, despite competitive iteration efficiency. The author implementation in TensorFlow Probability’s discussion directory uses JAX/FunMC; its location does not make it an installed TensorFlow transition kernel.

1.4.3 Generalized HMC and ensemble adaptation

Generalized HMC retains part of its momentum between transitions. One valid construction composes OU refreshment with the involutive leapfrog proposal $R\Phi$, accepts or rejects against H , and then applies an unconditional momentum flip R . The final flip preserves the joint target. In this equivalent convention, an accepted move ends with the forward momentum, whereas a rejected move flips it. Repeated rejections can therefore undo directional progress; correct momentum handling is part of the algorithm, not a cosmetic implementation detail.

MEADS adds ensemble-based scale adaptation and a persistent slice construction [Hoffman and Sountsov, 2022]. A slice variable has joint unnormalized density

$$\rho(x, p, s) = \mathbf{1}\{0 < s < e^{-H(x, p)}\}. \quad (1.43)$$

Integrating over s gives $e^{-H(x, p)}$. A volume-preserving reversible proposal at fixed s can move only within the slice; a valid update of s must preserve this joint measure. Reusing an arbitrary uniform variate across ordinary MH tests is not sufficient. The paper supplies a particular persistent-slice transition, together with ensemble estimates that avoid direct Hessian evaluation.

The conditional-independence argument behind folded ensemble adaptation can be stated precisely. Partition chains into a block A and its complement. Choose parameters

$\lambda = \Lambda(X_{-A})$ only from the complementary block. Conditional on X_{-A} , suppose the transition P_λ for A preserves $\pi^{\otimes|A|}$. Then

$$\int \pi^{\otimes|A|}(dx_A) \pi^{\otimes|A^c|}(dx_{-A}) P_{\Lambda(x_{-A})}(x_A, dx'_A) \quad (1.44)$$

$$= \pi^{\otimes|A|}(dx'_A) \pi^{\otimes|A^c|}(dx_{-A}). \quad (1.45)$$

The product target is preserved. Sequentially changing the held-out block preserves it as well. If the parameter estimate uses the very state being updated, the displayed conditional argument no longer applies. MEADS's fold construction must therefore be preserved when claiming its exact ensemble-adaptation property. For a first local implementation, finite adaptation followed by a completely frozen kernel is simpler to validate.

1.5 Microcanonical dynamics and the MAMS correction

1.5.1 A velocity of fixed norm

MAMS uses a direction u on the unit sphere S^{d-1} instead of a Gaussian momentum [Robnik et al., 2025]. For $d > 1$, define

$$\dot{x} = u, \quad \dot{u} = -\frac{1}{d-1}(I - uu^\top)g(x), \quad g(x) = \nabla U(x). \quad (1.46)$$

The projected force changes direction while preserving speed. Indeed,

$$\frac{d}{dt}\|u\|^2 = -\frac{2}{d-1}\{u^\top g - \|u\|^2 u^\top g\} = 0 \quad \text{when } \|u\| = 1. \quad (1.47)$$

The factor $d-1$ is the dimension of the tangent space. It makes this formula undefined at $d=1$; a one-dimensional specialization requires a different construction, not an arbitrary division guard.

The name microcanonical can obscure an important distinction: these equations are not ordinary canonical Hamiltonian equations in (x, u) with Gaussian kinetic energy. They are a time-rescaled, fixed-speed construction with stationary measure

$$\rho(x, u) dx d\omega(u) = \pi(x) dx \frac{d\omega(u)}{|S^{d-1}|}, \quad (1.48)$$

where $d\omega$ is sphere surface measure. The measure, and not merely the dimension of an array storing u , determines the Jacobian in the acceptance calculation.

1.5.2 Derivation of the stationary measure

For fixed $a \in \mathbb{R}^d$, the tangent field $v(u) = (I - uu^\top)a$ has intrinsic divergence

$$\operatorname{div}_{S^{d-1}} v = -(d-1)u^\top a. \quad (1.49)$$

To see this without a coordinate chart, let $P = I - uu^\top$ and use an orthonormal tangent frame. The ambient derivative is $Dv = -ua^\top - (u^\top a)I$. Taking the tangent trace gives $\operatorname{tr}(PDvP) = -(u^\top a)\operatorname{tr}(P) = -(d-1)u^\top a$, since $Pu = 0$. With $a = -g/(d-1)$, the

velocity field in (1.46) therefore has divergence $u^\top g$. Its position component has zero position divergence. The stationary continuity equation is satisfied because

$$\operatorname{div}(\rho b) = \rho\{\operatorname{div} b + b \cdot \nabla \log \rho\} \quad (1.50)$$

$$= \rho\{u^\top g - u^\top g\} = 0. \quad (1.51)$$

This derivation assumes sufficient smoothness and a nonexplosive flow for the continuity argument. It establishes the invariant density of the exact dynamics. It does not prove ergodicity of the deterministic motion or remove finite-step integration bias [Robnik et al., 2025, Appendix B.4].

The comparison with canonical HMC is useful. In canonical coordinates the flow preserves volume and density separately through energy conservation. Here density changes along the position trajectory and volume changes in the sphere directions compensate. The numerical method must keep track of that compensation.

1.5.3 The exact force subflow

Split the dynamics into a drift $A_h : (x, u) \mapsto (x + hu, u)$ and a force subflow B_h holding x fixed. For nonzero g , put

$$e = -g/\|g\|, \quad k = \|g\|/(d-1), \quad c_0 = e^\top u_0, \quad \delta = kh. \quad (1.52)$$

Then $\dot{u} = k(e - (e^\top u)u)$. The scalar projection $c = e^\top u$ satisfies $\dot{c} = k(1 - c^2)$. Integrating $dc/(1 - c^2) = k dt$ gives

$$c_h = \frac{\sinh \delta + c_0 \cosh \delta}{\cosh \delta + c_0 \sinh \delta}. \quad (1.53)$$

For $|c_0| < 1$ this follows from the addition formula for tanh; the formula also has well-defined limits at $c_0 = \pm 1$. Write $r = u - ce$, with $e^\top r = 0$. The transverse equation is $\dot{r} = -kcr$. Define $D_h = \cosh \delta + c_0 \sinh \delta$. Since $d \log D_t/dt = kc_t$, integration yields $r_h = r_0/D_h$. Combining the parallel and perpendicular components gives the closed form

$$B_h(u_0) = \frac{u_0 + e\{\sinh \delta + (\cosh \delta - 1)c_0\}}{D_h}. \quad (1.54)$$

The denominator is positive for finite δ and $c_0 \in [-1, 1]$, since $\cosh \delta > |\sinh \delta|$. A direct norm check uses

$$1 - c_0^2 + (\sinh \delta + c_0 \cosh \delta)^2 = (\cosh \delta + c_0 \sinh \delta)^2 = D_h^2. \quad (1.55)$$

Thus the update remains on the sphere. At $g = 0$, the correct limiting update is $B_h(u) = u$; constructing e by dividing by zero is unnecessary.

1.5.4 The surface Jacobian and accumulated work

During B_h , the sphere divergence is $u^\top g = -(d-1)kc$. Liouville's formula on the sphere and the expression for D_h imply

$$\log J_{B_h}(u_0) = \int_0^h \operatorname{div}_{S^{d-1}} \dot{u}_t dt \quad (1.56)$$

$$= -(d-1) \int_0^h kc_t dt = -(d-1) \log D_h. \quad (1.57)$$

This is the negative of the paper’s quantity $\Delta K = (d - 1) \log D_h$ [Robnik et al., 2025, equation (8)]. Here ΔK denotes a volume correction; it is not a change in $\|u\|^2/2$, which is constant.

A symmetric composition such as $B_{h/2}A_hB_{h/2}$ is reversible under $R(x, u) = (x, -u)$. The drift is volume preserving, while force substeps contribute (1.57). Proposition 1.1 therefore yields

$$\log r = -U(x') + U(x) + \sum_j \log J_{B_j}, \quad (1.58)$$

$$W = U(x') - U(x) + (d - 1) \sum_j \log D_j, \quad (1.59)$$

$$a = \min\{1, e^{-W}\}. \quad (1.60)$$

Potential changes telescope across drift substeps. The signs matter: subtracting rather than adding ΔK in W would produce the wrong accepted measure. The inspected BlackJAX implementation accumulates this integrator work and adds the endpoint log-density difference with precisely this sign.

For numerical stability, the equivalent identity

$$D_h = \tfrac{1}{2}\{(1 + c_0)e^\delta + (1 - c_0)e^{-\delta}\} \quad (1.61)$$

permits a log-sum-exp calculation of $\log D_h$, including limiting zero weights when $c_0 = \pm 1$. The vector update needs a correspondingly stable implementation for large $|\delta|$. A guard that merely clips δ changes the map and its Jacobian; it cannot silently retain the original correction. Reversibility, surface Jacobian and zero-gradient limits should be tested together.

1.5.5 Sphere refreshment and the stochastic variant

A full refresh is $u = \xi/\|\xi\|$ for $\xi \sim \mathcal{N}(0, I)$; rotational invariance gives the uniform sphere law. A partial refresh used in microcanonical implementations has the form [Robnik and Seljak, 2025, Robnik et al., 2025]

$$u' = \frac{u + \eta\xi}{\|u + \eta\xi\|}, \quad \eta > 0. \quad (1.62)$$

At a fixed state-independent η , its transition density with respect to surface measure is

$$q(u' | u) = (2\pi\eta^2)^{-d/2} \int_0^\infty r^{d-1} \exp\left[-\frac{r^2 + 1 - 2ru^\top u'}{2\eta^2}\right] dr. \quad (1.63)$$

This follows by writing a Gaussian draw in polar coordinates. The expression depends only on $u^\top u'$, so $q(u' | u) = q(u | u')$ and the uniform sphere law is reversible. The zero-noise limit is the identity transition. In a symmetric stochastic path, these refreshment ratios cancel; the deterministic subflow Jacobians still accumulate. The extended-path argument in MAMS’s Appendix A handles the actual noise and rejection conventions. State-dependent refreshment strengths would require rechecking the ratio.

MAMS consequently has several controls: a geometry or preconditioner, epsilon, a trajectory length and, in its Langevin variant, a decoherence scale. The paper uses automatic adaptation, including a preliminary preconditioner estimate—in practice from

NUTS—and length selection using autocorrelation information. Its practical acceptance target is 0.90, with 0.99 used for its funnel experiment; these are author settings, not universal optimal values. The paper calls physical trajectory length L . This chapter uses τ for physical length and L for an integer integration count, with $\tau \approx L\epsilon$, to avoid confusion with BayesFilter’s tuning interface.

1.6 Unadjusted bias and the LAPS warm-up strategy

1.6.1 What a finite-step unadjusted chain samples

An exact invariant diffusion or deterministic flow does not imply that its finite-step discretization has the same invariant distribution. A Gaussian calculation exposes the distinction without asymptotic notation. Use the unit-mass leapfrog in (1.15) for $U(x) = x^2/(2\sigma^2)$, refresh momentum independently before each single step, and omit MH. With $h = \epsilon$ and $y = h^2/\sigma^2$, its matrix is

$$\begin{pmatrix} x' \\ p' \end{pmatrix} = \begin{pmatrix} 1 - y/2 & h \\ -h\sigma^{-2}(1 - y/4) & 1 - y/2 \end{pmatrix} \begin{pmatrix} x \\ p \end{pmatrix}. \quad (1.64)$$

For $0 < y < 4$ the position recursion has coefficient $a = 1 - y/2$ with $|a| < 1$. If its stationary variance is v_h , then $v_h = a^2 v_h + h^2$. Solving gives

$$v_h = \frac{h^2}{1 - a^2} = \frac{\sigma^2}{1 - y/4}, \quad b = \frac{v_h}{\sigma^2} - 1 = \frac{y/4}{1 - y/4}. \quad (1.65)$$

Every positive stable step size inflates the variance in this scheme. Increasing the number of iterations only estimates v_h more accurately; it does not make v_h equal to σ^2 . Different splitting conventions can have different bias, so their formulas must not be exchanged silently.

The energy error contains useful information about this bias. Let A be the matrix in (1.64), $D = \text{diag}(\sigma^{-2}, 1)$ and $S = \text{diag}(v_h, 1)$. At the stationary unadjusted input, $z = (x, p) \sim \mathcal{N}(0, S)$, and

$$\Delta H = \frac{1}{2} z^\top Q z, \quad Q = A^\top D A - D. \quad (1.66)$$

For a centered Gaussian vector, expansion using fourth moments gives $\text{Var}(z^\top Q z) = 2 \text{tr}(Q S Q S)$. Hence

$$\text{Var}(\Delta H) = \frac{1}{2} \text{tr}(Q S Q S) = \frac{y^3}{16(1 - y/4)} \quad (1.67)$$

$$= \frac{4b^3}{(1 + b)^2} = F(b^2), \quad F(D) = \frac{4D^{3/2}}{(1 + \sqrt{D})^2}. \quad (1.68)$$

For an explicit check, substitution into Q gives

$$Q_{11} = -\frac{y^2(1 - y/4)}{4\sigma^2}, \quad Q_{12} = Q_{21} = \frac{hy(1 - y/2)}{4\sigma^2}, \quad Q_{22} = \frac{y^2}{4}. \quad (1.69)$$

Consequently $Q_{11}v_h = -y^2/4$ and $\frac{1}{2} \text{tr}(Q S Q S) = y^4/16 + y^3(1 - y/2)^2/[16(1 - y/4)]$. Combining the numerator uses $y(1 - y/4) + (1 - y/2)^2 = 1$ and yields (1.68). This is a

relation between the stationary error of this specified integrator and a specified covariance bias, not a relation between arbitrary transient error and arbitrary observables.

For independent Gaussian eigendirections, independent energy errors add. With $b_i = v_{h,i}/\sigma_i^2 - 1$, define

$$D_{\text{cov}} = \frac{1}{d} \sum_i b_i^2, \quad E = \frac{1}{d} \text{Var}(\Delta H) = \frac{1}{d} \sum_i F(b_i^2). \quad (1.70)$$

In the isotropic case $E = F(D_{\text{cov}})$ exactly. More generally,

$$F'(D) = \frac{2\sqrt{D}(3 + \sqrt{D})}{(1 + \sqrt{D})^3} > 0, \quad (1.71)$$

$$F''(D) = \frac{3 - 4\sqrt{D} - D}{\sqrt{D}(1 + \sqrt{D})^4}. \quad (1.72)$$

The numerator of the second derivative is positive for $0 < D < 11 - 4\sqrt{7}$. If every b_i^2 lies in that interval, Jensen's inequality gives $F(D_{\text{cov}}) \leq E$, and monotonicity implies $D_{\text{cov}} \leq F^{-1}(E)$. This is a sufficient restricted Gaussian bound. The broader statement and its admissible range in Robnik et al. [2026, Theorem 4.2 and Appendix B] require their stated conditions. Neither argument proves a universal microcanonical or non-Gaussian error bound.

For small h , $b = O(h^2)$, $D_{\text{cov}} = O(h^4)$ and $E = O(h^6)$. Thus a sixth-root adjustment $h_{\text{new}} = h(E_{\text{wanted}}/E)^{1/6}$ has a local scaling rationale. Outside that regime, the relation need not be accurate. The cited paper's non-Gaussian experiments include a Brownian-motion case where the proposed bound underpredicts observed bias by about a factor of 1.5. Its June 2026 revision provides useful evidence for adaptation, not a general certificate of approximation error.

1.6.2 The ideal bias–contraction schedule

LAPS uses an unadjusted microcanonical transient, followed by adjusted MAMS [Robnik and Seljak, 2026]. Its theoretical motivation can be reconstructed from two assumptions. Let D now denote a genuine distance satisfying the triangle inequality. Assume K_h has invariant distribution π_h , with

$$D(\pi_h, \pi) \leq C_b h^\kappa, \quad (1.73)$$

$$D(\rho K_h, \pi_h) \leq (1 - c)D(\rho, \pi_h), \quad 0 < c < 1, \quad (1.74)$$

where the constants are uniform over the step sizes being used. Two triangle inequalities imply

$$D(\rho K_h, \pi) \leq D(\rho K_h, \pi_h) + D(\pi_h, \pi) \quad (1.75)$$

$$\leq (1 - c)D(\rho, \pi) + (2 - c)C_b h^\kappa. \quad (1.76)$$

This separates contraction of initialization error from the numerical bias floor. Suppose one transition simulates fixed physical length τ and therefore costs approximately τ/h gradient evaluations. If $D_t = D(\rho_t, \pi)$, the lower bound on improvement per gradient is

$$R(h) = \frac{h}{\tau} \{cD_t - (2 - c)C_b h^\kappa\}. \quad (1.77)$$

Differentiating gives $\tau R'(h) = cD_t - (\kappa + 1)(2 - c)C_b h^\kappa$. The positive maximizer consequently satisfies

$$C_b h_t^\kappa = \frac{cD_t}{(2 - c)(\kappa + 1)}, \quad D_{t+1} \leq \left(1 - \frac{c\kappa}{\kappa + 1}\right) D_t. \quad (1.78)$$

This reproduces the argument in LAPS Appendix A. The optimum is of the specified lower bound under uniform contraction and cost assumptions. It is not a universal optimum for arbitrary MCMC. In particular, the moment statistic used in practice below is not a general distance with the triangle inequality, so it cannot simply be substituted into this proof.

1.6.3 Equipartition as a measurable but incomplete diagnostic

The target moments needed to measure D_t are generally unknown. LAPS uses an integration-by-parts identity. Let ρ describe the current ensemble, with mean m_ρ . Define

$$V_{ij}(\pi, \rho) = \mathbb{E}_\rho[-(X_i - m_{\rho,i})\partial_j \log \pi(X)], \quad \tilde{D} = \frac{1}{d} \|I - V\|_F^2. \quad (1.79)$$

Assume the relevant expectations exist, π is differentiable and the boundary term $(x_i - m_{\pi,i})\pi(x)$ vanishes in the integration-by-parts calculation. At the target,

$$V_{ij}(\pi, \pi) = - \int (x_i - m_{\pi,i}) \partial_j \pi(x) dx \quad (1.80)$$

$$= - [\text{boundary term}] + \int \partial_j (x_i - m_{\pi,i}) \pi(x) dx = \delta_{ij}. \quad (1.81)$$

The identity uses the score of the actual density, including coordinate Jacobians. A generic force b cannot be inserted without a new argument. Bounded domains and heavy tails require particular care with the boundary term and existence of moments.

The implication $\rho = \pi \Rightarrow V = I$ does not have a general converse. For $\pi = \mathcal{N}(0, I)$, $-\nabla \log \pi(x) = x$. If $\rho = \mathcal{N}(m, S)$, then

$$V = \mathbb{E}_\rho[(X - m)X^\top] = S. \quad (1.82)$$

Thus $S = I$ makes the diagnostic zero for every mean m , however far it is from the target. The full matrix detects a wrong covariance, but its diagonal reduction

$$\tilde{D}_{\text{diag}} = \frac{1}{d} \sum_i (1 - V_{ii})^2 \quad (1.83)$$

does not. For instance, $S = \begin{pmatrix} 1 & r \\ r & 1 \end{pmatrix}$ with $0 < |r| < 1$ has zero diagonal diagnostic and wrong correlation. Even the full matrix cannot distinguish all non-Gaussian distributions: a centered smooth mixture $\frac{1}{2}\mathcal{N}(-a, 1 - a^2) + \frac{1}{2}\mathcal{N}(a, 1 - a^2)$, $0 < a < 1$, has variance one and therefore $V = 1$, while its fourth moment $3 - 2a^4$ differs from the standard Gaussian value three. These are exact counterexamples, not merely concerns about finite ensemble size.

The LAPS paper explicitly calls $\tilde{D}$ a proxy and proves separation only on a restricted zero-mean Gaussian family. The examples above explain why one needs additional means, dependence, tail and mode diagnostics when using it for adaptation. Empirical moment fluctuations can also stabilize inside a single missed mode. A stable statistic is informative about that statistic, not all posterior quantities.

1.6.4 From the proxy to an operational schedule

In the paper’s scheme, select a fraction $0 < C < 1$ and set

$$D_{\text{wanted}} = C\tilde{D}_t, \quad E_{\text{wanted}} = F(D_{\text{wanted}}), \quad h_{t+1} = h_t \left(\frac{E_{\text{wanted}}}{\hat{E}_t} \right)^{1/6}. \quad (1.84)$$

The first assignment expresses a desired numerical bias below the estimated current error; the second borrows the Gaussian relationship; the third uses the small-step energy-error scaling. Each link has assumptions. In particular, $\hat{E}_t$ is measured during a transient, whereas (1.68) was derived at an unadjusted stationary law. Finite ensembles, small denominators, heavy tails and a wrong proxy can all mislead the update. A bounded implementation must handle them explicitly and report why adaptation ended.

After the unadjusted stage, LAPS applies a diagonal reparameterization, turns on MH and adapts the final adjusted settings. The paper uses a practical stopping rule based on fluctuations of second moments, not a proof that all initialization bias has disappeared. For posterior reporting, discard every unadjusted and adaptive draw, freeze all shared settings and assess the final fixed kernel. If chains shared adaptation, freezing stops further coupling but does not instantly make their current states independent or stationary.

The inspected BlackJAX revision contains choices different from the paper. Its desired error is coded as $C \text{bias}^{3/8}$, with the paper’s $F(C\tilde{D})$ expression commented out. Its diagonal statistic uses $-x_i \partial_i \log \pi(x)$ without centering, and its fluctuation function uses variance divided by squared mean rather than relative standard deviation. The centered mean-shift counterexample therefore does not apply unchanged to that uncentered code, but neither statistic determines a general distribution. The adjusted epsilon freezes through an acceptance-tolerance flag; the first tolerance-satisfying update can still change epsilon before the next call freezes it. A faithful local port must resolve these definitions and test the setting actually frozen. These observations concern the pinned current source, not an identified source revision of the published benchmark.

1.6.5 Gradient work and parallel latency

Table 1.1 reports selected entries from the MAMS paper. The criterion is a normalized squared error in the worst second moment below 0.01, taking median error across at least 128 chains. Tuning work is excluded. A small ratio in one model does not invalidate the method, but it does limit a blanket several-fold claim.

Author target	NUTS gradients	MAMS gradients	Ratio
Gaussian	19,652	3,249	6.05
Banana	95,519	14,078	6.78
Bimodal	210,758	139,418	1.51
Brownian	29,816	13,528	2.20
Stochastic volatility	843,768	430,088	1.96

Table 1.1: Selected MAMS Table 1 entries [Robnik et al., 2025]. Ratios are descriptive arithmetic on the authors’ reported counts, not local timings or uncertainty-qualified rankings for the present consumers.

LAPS's experiments use an NVIDIA A100 with 40 GB. Its Table 1 uses 4,096 chains; Table 2 uses 256. For stochastic volatility, Table 2 reports 29,900 gradients for sequential NUTS, 1,100 per LAPS chain and 281,600 across the LAPS ensemble [Robnik and Seljak, 2026]. The last two figures satisfy $256 \times 1,100 = 281,600$. Under the idealized model of perfectly synchronized work, its latency ratio to that sequential comparator is

$$\frac{C_{\text{LAPS}}}{C_{\text{NUTS}}} \approx \frac{1,100 c_{\nabla}(256)}{29,900 c_{\nabla}(1)}. \quad (1.85)$$

Per-chain work is about 27 times smaller, but total gradient work is about 9.4 times larger. The idealized latency gain requires $c_{\nabla}(256)/c_{\nabla}(1) < 29,900/1,100$ before other overheads. This calculation is not a fair complete local benchmark by itself: a local NUTS baseline may also be batched, compilation and adaptation must be counted, and a requested MCSE may require more than one final draw per chain. The paper provides motivation to measure batch scaling, not permission to assume it for Kalman, UKF or QZ calculations.

1.7 Geometry and initialization

1.7.1 Affine whitening and nonlinear funnels

For a Gaussian target, let $\Sigma = AA^{\top} \in \mathbb{R}^{d \times d}$ be its symmetric positive-definite covariance, with A invertible. The map $x = \mu + Az$ gives a standard normal density for z . Substitution into (1.7) gives

$$-\frac{1}{2}(x - \mu)^{\top} \Sigma^{-1}(x - \mu) + \log |\det A| = -\frac{1}{2}z^{\top} z + \text{constant}. \quad (1.86)$$

For ordinary Gaussian HMC in the original coordinates, equal oscillation frequencies correspond to $M = \Sigma^{-1}$ in the convention $K(p) = p^{\top} M^{-1} p/2$. Libraries sometimes expose M^{-1} as their *inverse mass* parameter, so verbal descriptions of mass adaptation are ambiguous unless this convention is stated. Whitening and mass adaptation repair constant linear anisotropy. They cannot generally remove position-dependent conditional scales.

Consider the funnel

$$V \sim \mathcal{N}(0, s^2), \quad X_i | V = v \sim \mathcal{N}(0, e^v), \quad i = 1, \dots, m. \quad (1.87)$$

Its log density, up to a constant, is

$$-\frac{v^2}{2s^2} - \frac{m}{2}v - \frac{1}{2}e^{-v} \sum_i x_i^2. \quad (1.88)$$

The child-coordinate curvature is e^{-v} and becomes arbitrarily large in the neck. Let $v = sz_0$ and $x_i = e^{v/2} z_i$. The Jacobian is block triangular, with determinant $se^{mv/2}$. Adding its logarithm cancels the linear $-mv/2$ term and yields

$$\log \pi_z(z) = -\frac{1}{2}z_0^2 - \frac{1}{2} \sum_i z_i^2 + \text{constant}. \quad (1.89)$$

Thus exact noncentering solves this analytic geometry problem. It does not show that an affine tuning procedure ought to solve the centered funnel.

A partial-whitening example is equally useful. Use $x_i = e^{v/2-\delta(v)}z_i$ and retain v as a coordinate. Its transformed log density is

$$\log \pi(v, z) = -\frac{v^2}{2s^2} - m\delta(v) - \frac{1}{2}e^{-2\delta(v)} \sum_i z_i^2 + \text{constant}. \quad (1.90)$$

The child Hessian of the negative log density is $e^{-2\delta(v)}I$. If $|\delta(v)| \leq a$, its eigenvalues are bounded between e^{-2a} and e^{2a} . This bounds that block only: derivatives of δ create cross terms and curvature in v . It is wrong to infer a global full-Hessian bound from this child-coordinate bound. Exact and residual-map fixtures distinguish map correctness from robustness to remaining geometry.

1.7.2 What neural transport learns

For base density $\phi(z)$ and invertible F_λ , the induced density q_λ obeys $\log q_\lambda(F_\lambda(z)) = \log \phi(z) - \log |\det J_{F_\lambda}(z)|$. The reverse-KL objective is therefore

$$\text{KL}(q_\lambda \parallel \pi) = \mathbb{E}_{Z \sim \phi} [\log \phi(Z) - \log |\det J_{F_\lambda}(Z)| + U(F_\lambda(Z))] + \log Z_\pi. \quad (1.91)$$

The unknown constant $Z_\pi = \int e^{-U(x)} dx$ does not affect optimization. This is a training objective for useful coordinates, not a posterior-error certificate [Hoffman et al., 2019]. Undercoverage of a mode contributes little when that mode is rarely sampled by q_λ . The final transformed target must still use (1.7), not the learned approximation alone.

Once frozen, the same map can be used by ordinary HMC, SNAPER or MAMS. Consequently a fair kernel comparison holds the map fixed. An end-to-end comparison may let each method construct geometry, but must then count fitting, inverse-map evaluation for matched model starts, forward/log-Jacobian work and transformed gradients. The current BayesFilter architecture policy selects configured Huang DSF/NAF for the October 2026 forward/reverse training study, and the repaired author IAF outside that study. Architecture selection and successful training are separate from map validity, tail coverage and downstream sampling accuracy.

1.7.3 Pathfinder from a quadratic expansion

Pathfinder constructs Gaussian approximations along optimization paths [Zhang et al., 2022]. Let x_k be an L-BFGS iterate minimizing U , $g_k = \nabla U(x_k)$, and B_k a positive-definite inverse-Hessian approximation. A local quadratic model is

$$U(x) \approx U(x_k) + g_k^\top (x - x_k) + \frac{1}{2}(x - x_k)^\top B_k^{-1}(x - x_k). \quad (1.92)$$

Completing the square gives mean and covariance

$$\mu_k = x_k - B_k g_k, \quad q_k = \mathcal{N}(\mu_k, B_k). \quad (1.93)$$

In particular, the Gaussian is centered at the stationary point of the local quadratic, not necessarily the current iterate. If $s_k = x_{k+1} - x_k$ and $y_k = g_{k+1} - g_k$, the full BFGS inverse update is

$$B_{k+1} = (I - \rho_k s_k y_k^\top) B_k (I - \rho_k y_k s_k^\top) + \rho_k s_k s_k^\top, \quad \rho_k = (y_k^\top s_k)^{-1}. \quad (1.94)$$

Multiplying by y_k verifies the secant equation $B_{k+1}y_k = s_k$. If $y_k^\top s_k > 0$ and B_k is positive definite, the quadratic form of the update is the sum of a nonnegative transformed B_k form and $\rho_k(s_k^\top v)^2$; their simultaneous vanishing forces $v = 0$. This proves positive definiteness. Limited-memory representations retain selected (s, y) pairs and exploit a diagonal-plus-low-rank form instead of storing a dense $d \times d$ matrix. Negative or negligible curvature must be handled explicitly, not treated as a valid Gaussian covariance.

For each proposed q_k , the evidence lower bound satisfies

$$\mathcal{L}(q_k) = \mathbb{E}_{q_k}[\log \tilde{\pi}(X) - \log q_k(X)] = \log Z_\pi - \text{KL}(q_k \parallel \pi). \quad (1.95)$$

Maximizing an estimate of $\mathcal{L}$ along the path chooses a promising approximation; the estimated maximum has Monte Carlo selection uncertainty. The inspected Stan implementation checks curvature and stores the approximation with favorable estimated ELBO. A good local approximation can provide initial states and scaling, but missing modes remain possible.

The paper’s multipath importance step is best understood on an augmented space. If J paths supply densities q_j , define

$$q(x, j) = J^{-1}q_j(x), \quad \pi(x, j) = J^{-1}\pi(x). \quad (1.96)$$

The importance ratio is $\tilde{\pi}(x)/q_j(x)$ for a draw from path j ; the factors J^{-1} cancel. This avoids evaluating every mixture component at every draw. It is different from the marginal-mixture ratio $\tilde{\pi}(x)/[J^{-1}\sum_j q_j(x)]$, although both can define proper importance schemes with appropriate sampling. Pathfinder applies Pareto-smoothed importance resampling on the augmented construction. Finite resampling is an approximation; importance-weight concentration and tail diagnostics matter. It should be evaluated as an initializer before being credited with a reduction in total MCMC work.

1.8 Alternatives for residual stiffness and discontinuities

1.8.1 Delayed rejection with the reverse path included

If one global epsilon must be tiny in a small stiff region, delayed rejection can attempt a smaller step after a rejected proposal [Turok et al., 2026]. The principle is easiest to derive for ordinary proposal densities. First propose $y \sim q_1(x, \cdot)$ and accept with $\alpha_1(x, y)$. After rejection, propose $z \sim q_2(x, y, \cdot)$. The forward probability of this two-stage path, before the second decision, is $\pi(x)q_1(x, y)(1 - \alpha_1(x, y))q_2(x, y, z)$. Its reverse counterpart starts from z , rejects y , and proposes x . Consequently the second-stage acceptance is

$$\alpha_2(x, y, z) = 1 \wedge \frac{\pi(z)q_1(z, y)[1 - \alpha_1(z, y)]q_2(z, y, x)}{\pi(x)q_1(x, y)[1 - \alpha_1(x, y)]q_2(x, y, z)}. \quad (1.97)$$

Multiplying either path density by its acceptance gives the minimum of the two path densities, proving equality of accepted flows. The factors for earlier rejections are essential. Halving epsilon after a rejection and applying only the ordinary endpoint MH ratio drops them.

Deterministic GHMC proposals require the corresponding reversible-map construction rather than pretending their transition laws have ordinary Lebesgue densities. The

author implementation evaluates reverse *ghost* proposals to obtain the earlier rejection probabilities. Recursion can make work grow exponentially with retry depth. A bounded depth, evaluation counts including ghost trajectories, and forward/reverse tests are therefore part of a meaningful local comparison. Author experiments with reference-draw initialization do not establish cold-start performance on economic posteriors.

1.8.2 A true potential jump is not a steep smooth gradient

For a coordinate with Laplace kinetic energy $K(p) = |p|/m$, its speed away from $p = 0$ is $\dot{x} = \text{sign}(p)/m$. At a boundary with potential jump $\Delta U = U_+ - U_-$, conservation of energy requires

$$|p_+| = |p_-| - m\Delta U. \quad (1.98)$$

If the right side is positive, the trajectory transmits with the same direction and reduced kinetic energy for an upward jump. Otherwise it reflects, reversing momentum while staying on the initial side. A downward jump increases kinetic energy. This describes the local physical rule behind Laplace-momentum discontinuous HMC; the complete numerical proposal also needs its reversible coordinate update and appropriate treatment of mixed smooth coordinates [Nishimura et al., 2020].

Changing NUTS to MAMS does not by itself implement those events. A DSGE regime boundary might instead create a discontinuous score of a continuous density, a discontinuity in the density, or an invalid region. These are different mathematical cases. Their target and boundary behavior must be determined before selecting a discontinuous method. This specialist option was examined at the paper level; no claim of an audited TensorFlow port follows.

1.9 State-space structure and likelihood cost

1.9.1 Associative filtering by unnormalized kernels

For fixed parameters, write a filtering factor as $f_i(x | z)g_i(z)$, where f_i is a conditional state density and g_i is the likelihood of the observations in that segment given its input state. Combine consecutive factors by

$$Z_{ij}(z) = \int g_j(y)f_i(y | z) dy, \quad (1.99)$$

$$g_{ij}(z) = g_i(z)Z_{ij}(z), \quad (1.100)$$

$$f_{ij}(x | z) = \frac{\int f_j(x | y)g_j(y)f_i(y | z) dy}{Z_{ij}(z)}. \quad (1.101)$$

Assume the integrals exist and the normalizing denominator is positive. Multiplication eliminates the normalization:

$$g_{ij}(z)f_{ij}(x | z) = \int [g_j(y)f_j(x | y)][g_i(z)f_i(y | z)] dy. \quad (1.102)$$

If $K_i(x, z) = g_i(z)f_i(x | z)$, the right side is kernel convolution. For three segments, both parenthesizations integrate $K_k(x, y_2)K_j(y_2, y_1)K_i(y_1, z)$ over (y_1, y_2) . Tonelli's theorem for nonnegative kernels, or Fubini under absolute integrability, permits the order to be exchanged. The unnormalized operation is therefore associative; the uniquely normalized

(f, g) pair inherits that property. This is the foundation of temporal parallel filtering [Särkkä and García-Fernández, 2020, Definition 2 and Theorem 3].

An associative scan computes every prefix. A balanced evaluation has $O(\log T)$ dependency depth and $O(T)$ total binary combinations. It needs enough processors and memory for the levels of the tree. Its arithmetic constants may be larger than those of a sequential Kalman recursion. Thus logarithmic depth is not logarithmic total work or a guarantee of speedup.

1.9.2 Gaussian closure and the likelihood constant

Gaussian factors make the convolution tractable. Let r collect the retained variables at a join and let $y \in \mathbb{R}^m$ be the state being integrated out. Write the product of the two unnormalized factors in canonical form,

$$\exp\left\{a - \frac{1}{2}r^\top J_{rr}r - r^\top J_{ry}y - \frac{1}{2}y^\top J_{yy}y + h_r^\top r + h_y^\top y\right\}, \quad (1.103)$$

where $r \in \mathbb{R}^k$, $J_{rr} \in \mathbb{R}^{k \times k}$, $J_{ry} \in \mathbb{R}^{k \times m}$, $h_r \in \mathbb{R}^k$, and $h_y \in \mathbb{R}^m$. The block $J_{yy} \in \mathbb{R}^{m \times m}$ is symmetric positive definite and $J_{yr} = J_{ry}^\top$. Completing the square in y gives

$$\bar{J} = J_{rr} - J_{ry}J_{yy}^{-1}J_{yr}, \quad (1.104)$$

$$\bar{h} = h_r - J_{ry}J_{yy}^{-1}h_y, \quad (1.105)$$

$$\bar{a} = a + \frac{m}{2} \log(2\pi) - \frac{1}{2} \log \det J_{yy} + \frac{1}{2} h_y^\top J_{yy}^{-1} h_y. \quad (1.106)$$

For verification, write $b_y = h_y - J_{yr}r$ and expand $\frac{1}{2}b_y^\top J_{yy}^{-1}b_y$ after integrating the centered Gaussian. The quadratic, linear and constant terms are exactly those above. Dropping $\bar{a}$ retains a proportional state density but loses the likelihood normalization required by parameter HMC. Deterministic state components may make a naive precision block singular; the formula then needs an appropriate constrained or square-root representation. Adding an undeclared ridge changes the target.

The serial Gaussian likelihood increment is

$$\ell_t = -\frac{1}{2}\{n_t \log(2\pi) + \log \det S_t + v_t^\top S_t^{-1}v_t\}, \quad (1.107)$$

where $v_t \in \mathbb{R}^{n_t}$ is the innovation and $S_t \in \mathbb{R}^{n_t \times n_t}$ is its symmetric positive-definite covariance. For a fixed observation pattern its differential is

$$d\ell_t = -\frac{1}{2} \text{tr}(S_t^{-1}dS_t) - v_t^\top S_t^{-1}dv_t + \frac{1}{2}v_t^\top S_t^{-1}(dS_t)S_t^{-1}v_t. \quad (1.108)$$

This follows from $dS^{-1} = -S^{-1}(dS)S^{-1}$ and $d \log \det S = \text{tr}(S^{-1}dS)$. The differentials include the recursive dependence of innovations and covariances on parameters. Agreement of filter values alone does not establish agreement of these total derivatives.

1.9.3 Correlated noises and missing components

Consider a transition and measurement

$$X' = AX + b + w, \quad Y = HX' + d + v, \quad \begin{pmatrix} w \\ v \end{pmatrix} \sim \mathcal{N}\left(0, \begin{pmatrix} Q & C \\ C^\top & R \end{pmatrix}\right). \quad (1.109)$$

Conditionally on $X = x$, the mean of Y is $H(Ax + b) + d$, its covariance is

$$S = HQH^\top + HC + C^\top H^\top + R, \quad \text{Cov}(X', Y | X) = QH^\top + C. \quad (1.110)$$

The conditional Gaussian formula therefore yields

$$K = (QH^\top + C)S^{-1}, \quad (1.111)$$

$$X' \mid X = x, Y = y \sim \mathcal{N}((I - KH)(Ax + b) + K(y - d), Q - KSK^\top). \quad (1.112)$$

This supplies the f factor; $g(x)$ is the Gaussian density of y with the preceding mean and covariance S . Correlated noise does not destroy the abstract associative construction, but an implementation written only for independent noises must be extended through an equivalent formulation. If a matrix O selects the observed components, replace y, H, d, C, R by $Oy, OH, Od, CO^\top, ORO^\top$. An entirely missing observation instead has $g = 1$ and leaves the transition factor unchanged. Whole-time masks and component-specific masks are therefore different interfaces.

TensorFlow Probability exposes a parallel LGSSM option and a parallel-filter library. The inspected source returns likelihoods and supports time/batch masks. A direct replacement of MacroFinance's correlated-noise, component-masked recursion is not established by that source availability. Test the algebraic reformulation, likelihood, total score, dtype, compilation and memory behavior on the actual target. For nonlinear filters whose linearization or sigma points depend recursively on the current filtering law, an independently parallelized linear Gaussian scan is not automatically the same approximation.

1.9.4 Conditional particle methods for latent paths

If the unknown includes a long path, write its fixed-parameter smoothing density as

$$\pi_T(x_{1:T}) = Z_T^{-1} \gamma_T(x_{1:T}), \quad \gamma_T = M_1(x_1)G_1(x_1) \prod_{t=2}^T M_t(x_t \mid x_{t-1})G_t(x_{t-1}, x_t). \quad (1.113)$$

Assume $0 < Z_T < \infty$ and proposal support covering the target. Ordinary particle filtering with N particles uses proposal r_1 and r_t , samples ancestors with normalized weights W_{t-1} , and assigns incremental weights

$$w_1^i = \frac{M_1(x_1^i)G_1(x_1^i)}{r_1(x_1^i)}, \quad w_t^i = \frac{M_t(x_t^i \mid x_{t-1}^{a_t^i})G_t(x_{t-1}^{a_t^i}, x_t^i)}{r_t(x_t^i \mid x_{t-1}^{a_t^i})}. \quad (1.114)$$

The particle-generation law is

$$\psi(\mathbf{x}, \mathbf{a}) = \prod_{i=1}^N r_1(x_1^i) \prod_{t=2}^T \prod_{i=1}^N W_{t-1}^{a_t^i} r_t(x_t^i \mid x_{t-1}^{a_t^i}). \quad (1.115)$$

For a selected final index k , define its ancestry by $b_T = k$ and $b_{t-1} = a_t^{b_t}$. The extended target

$$\Phi(\mathbf{x}, \mathbf{a}, k) = \frac{\pi_T(x_{1:T}^{b_{1:T}})}{N^T} \frac{\psi(\mathbf{x}, \mathbf{a})}{r_1(x_1^{b_1}) \prod_{t=2}^T W_{t-1}^{b_{t-1}} r_t(x_t^{b_t} \mid x_{t-1}^{b_{t-1}})} \quad (1.116)$$

has the desired selected-path marginal under the usual support conditions. To see the marginal, fix the selected path and its index sequence. The remaining quotient is the conditional particle simulation law for all off-path variables; integrating them gives one.

Summing the N^T possible index sequences cancels N^{-T} . Conversely, holding the particle system fixed, substitute (1.114). The ratio of path factors is

$$\frac{\gamma_T(x^b)}{r_1(x_1^{b_1}) \prod_{t=2}^T W_{t-1}^{b_{t-1}} r_t(x_t^{b_t} | x_{t-1}^{b_{t-1}})} = w_T^k \prod_{t=1}^{T-1} \sum_i w_t^i. \quad (1.117)$$

All intermediate selected weights cancel, so the conditional law of k is W_T^k . Conditional particle simulation retaining the current path, followed by this terminal selection, is a Gibbs construction on Φ and leaves π_T invariant [Andrieu et al., 2010, Sections 4.2–4.5, equation (31)]. Additional backward sampling or ancestor updates must preserve the same extended target. More particles affect mixing and cost, not the identity of that marginal.

1.9.5 Particle gradients and auxiliary marginalization

Particle-MALA and Particle-mGRAD adapt this construction to local, gradient-informed moves [Corenflos and Finke, 2024, Algorithms 3–7]. The key is to introduce auxiliary variables with known conditional law and correct the particle weights for the augmented target. In schematic form,

$$\bar{\pi}(x, u) = \pi_T(x) q(u | x). \quad (1.118)$$

First draw $u | x$, then use a conditional particle kernel invariant for $\bar{\pi}(x | u)$. Both conditional updates preserve $\bar{\pi}$; integrating u returns π_T . A gradient shift in q guides proposals, but its density must also appear in the conditional target and weights. Changing proposals while retaining bootstrap weights is wrong.

The Gaussian calculation underlying the GRAD family is explicit. Suppose the transition at a fixed parent is $X \sim \mathcal{N}(m, Q)$ and an auxiliary observation has density $u | X \sim \mathcal{N}(X, rI)$, $r > 0$, with $Q \in \mathbb{R}^{d \times d}$ symmetric positive definite. Completing the square gives

$$V = (Q^{-1} + r^{-1}I)^{-1} = rA, \quad A = Q(Q + rI)^{-1}, \quad (1.119)$$

$$\mu = V(Q^{-1}m + r^{-1}u) = (I - A)m + Au. \quad (1.120)$$

If the reference state is x and $u \sim \mathcal{N}(x + r\nabla \log G(x), rI)$, averaging the conditional proposal over this auxiliary draw gives

$$X' \sim \mathcal{N}((I - A)m + A[x + r\nabla \log G(x)], r(A + AA^\top)). \quad (1.121)$$

The covariance follows from the law of total variance: V from the conditional draw and $ArIA^\top$ from its random mean. For small prior covariance relative to rI , $A \approx Q/r$ and the proposal approaches the prior transition. For weak prior information, $A \approx I$ and it approaches $\mathcal{N}(x + r\nabla \log G(x), 2rI)$, a Langevin proposal. This explains the interpolation between prior-guided particle moves and MALA-like moves.

The conditional importance correction can also be derived explicitly. Denote the conditional Gaussian proposal with mean μ and covariance V in (1.120) by $r_u(x)$, set $g(x) = \nabla \log G(x)$, and let the actual auxiliary density be $q(u | x) = \mathcal{N}(u; x + rg(x), rI)$. The Gaussian product identity is

$$\mathcal{N}(x; m, Q) \mathcal{N}(u; x, rI) = \mathcal{N}(u; m, Q + rI) r_u(x). \quad (1.122)$$

The target conditional on u contains $\mathcal{N}(x; m, Q)G(x)q(u \mid x)$. Dividing by $r_u(x)$ and using the identity gives the unnormalized weight

$$w(x; u) = G(x)\mathcal{N}(u; m, Q + rI) \frac{\mathcal{N}(u; x + rg(x), rI)}{\mathcal{N}(u; x, rI)} \quad (1.123)$$

$$= G(x)\mathcal{N}(u; m, Q + rI) \exp\{g(x)^\top(u - x) - \frac{r}{2}\|g(x)\|^2\}. \quad (1.124)$$

The exponent follows by expanding $-\{\|u - x - rg(x)\|^2 - \|u - x\|^2\}/(2r)$. The normalizing factor depends on the parent through m and must be retained when comparing ancestors. This calculation gives a complete local auxiliary correction, and explains why guidance cannot be added without adjusting the weights.

This one-particle marginal does not supply the complete Particle-mGRAD weights: particles can share auxiliary variables, so their joint law must also be marginalized. The paper derives those joint weight corrections. Its particular Particle-mGRAD construction assumes the transition covariance Q_t does not depend on the parent state; its more general auxiliary method allows parent-dependent covariance. The invariant distribution is the fixed-parameter smoothing law, not automatically the joint posterior in structural parameters and states. Theoretical interpolation results and high-dimensional stochastic-volatility examples support investigating it when latent paths are sampled. They do not establish an improvement over marginalized Kalman likelihood HMC. The inspected author implementation is JAX, and a TensorFlow implementation would be a separate project.

1.10 Warm-up, convergence and precision

Let $h(x)$ be a reported quantity: a structural parameter, an impulse response at a specified horizon, a predictive spread, or an indicator of an economically important event. For $\mu = \mathbb{E}_\pi h$, any estimator obeys

$$\mathbb{E}[(\hat{\mu} - \mu)^2] = (\mathbb{E}\hat{\mu} - \mu)^2 + \text{Var}(\hat{\mu}). \quad (1.125)$$

More independent chains can reduce the variance without removing a common initialization bias. An unadjusted integrator can introduce a stationary bias that survives an arbitrarily long run. A wrong likelihood changes μ itself. These errors cannot be diagnosed by a single acceptance statistic. Warm-up is intended to reduce initialization effects and learn a useful kernel; a Metropolis correction addresses the invariant distribution of a specified proposal; retained-sample diagnostics assess remaining disagreement and Monte Carlo precision. None implies the others.

1.10.1 Long-run variance and the Monte Carlo standard error

First suppose the retained chain is stationary. Write $Y_t = h(X_t) - \mu$ and $\gamma_k = \text{Cov}(Y_0, Y_k)$. Directly expanding the variance of the sum gives

$$\text{Var}\left(\frac{1}{n} \sum_{t=1}^n h(X_t)\right) = \frac{1}{n^2} \sum_{s=1}^n \sum_{t=1}^n \text{Cov}(Y_s, Y_t) \quad (1.126)$$

$$= \frac{1}{n} \left[\gamma_0 + 2 \sum_{k=1}^{n-1} \left(1 - \frac{k}{n}\right) \gamma_k \right]. \quad (1.127)$$

If the autocovariances are absolutely summable, the bracket converges to

$$\sigma_h^2 = \gamma_0 + 2 \sum_{k=1}^{\infty} \gamma_k, \quad \tau_h = \frac{\sigma_h^2}{\gamma_0}, \quad n_{\text{eff},h} = \frac{n}{\tau_h}. \quad (1.128)$$

For $\gamma_0 > 0$, τ_h is the integrated autocorrelation time. Negative correlations can make it less than one. A suitable Markov-chain central limit theorem supplies the stronger distributional conclusion $\sqrt{n}(\hat{\mu} - \mu) \Rightarrow \mathcal{N}(0, \sigma_h^2)$; absolute summability alone is not a general proof of that theorem. The estimated Monte Carlo standard error is $\widehat{\text{MCSE}}_h = \sqrt{\widehat{\sigma}_h^2/n}$. For a vector h , replace γ_k by $\Gamma_k = \text{Cov}(Y_0, Y_k)$ and set $\Sigma = \Gamma_0 + \sum_{k \geq 1} (\Gamma_k + \Gamma_k^\top)$. Each reported linear combination has its own variance $a^\top \Sigma a/n$. An ESS number divorced from its observable conceals this dependence.

Independent retained chains can be combined by variance addition. If chain j contributes n_j draws with long-run variance σ_j^2 , the draw-weighted pooled mean has approximate variance

$$\text{Var}(\hat{\mu}_{\text{pool}}) \simeq \frac{\sum_j n_j \sigma_j^2}{(\sum_j n_j)^2}. \quad (1.129)$$

This expression assumes the means estimate the same stationary expectation and cross-chain covariances vanish. Concatenating interacting ensemble chains and calling them one stationary chain does not establish those assumptions.

1.10.2 Batch means, lugsail correction and their limits

For a vector of observables $h : \mathbb{R}^d \rightarrow \mathbb{R}^p$, take $n = ab$, with $a > 1$ nonoverlapping batches of length b , and define $\bar{h}_k = b^{-1} \sum_{t=(k-1)b+1}^{kb} h(X_t)$. The batch-means covariance estimator is

$$\hat{\Sigma}_b = \frac{b}{a-1} \sum_{k=1}^a (\bar{h}_k - \bar{h})(\bar{h}_k - \bar{h})^\top. \quad (1.130)$$

If a batch is long enough, its mean has covariance approximately Σ/b , which explains the factor b . Increasing the total run while keeping b fixed does not generally remove the truncation bias. Consistency results require increasing batch length and number of batches, together with appropriate moment, mixing or strong-invariance conditions [Flegal and Jones, 2010, Section 3]; the multivariate lugsail extension is developed by Vats and Flegal [2021].

For integer $r > 1$ dividing b and $0 \leq c < 1$, the lugsail construction is

$$\hat{\Sigma}_{\text{lug}} = \frac{\hat{\Sigma}_b - c \hat{\Sigma}_{b/r}}{1 - c}. \quad (1.131)$$

Its purpose is visible from the leading bias. Under the assumptions of Vats and Flegal [2021, Theorem 4], write $\mathbb{E} \hat{\Sigma}_b - \Sigma = \mathcal{G}/b + o(b^{-1})$. Substituting the expansion at both batch sizes gives

$$\mathbb{E} \hat{\Sigma}_{\text{lug}} - \Sigma = \frac{1}{1-c} \left(\frac{\mathcal{G}}{b} - c \frac{r\mathcal{G}}{b} \right) + o(b^{-1}) \quad (1.132)$$

$$= \frac{1 - rc}{1 - c} \frac{\mathcal{G}}{b} + o(b^{-1}). \quad (1.133)$$

For a scalar process with predominantly positive correlations, $\mathcal{G} = -2 \sum_{k \geq 1} k \gamma_k$ is often negative. The choice $rc > 1$ reverses this leading bias; $rc = 1$ cancels it. For example, a stationary AR(1) correlation $\gamma_k = \gamma_0 \phi^k$, $0 < \phi < 1$, gives

$$\sigma_h^2 = \gamma_0 \frac{1 + \phi}{1 - \phi}, \quad \mathcal{G} = -\frac{2\gamma_0 \phi}{(1 - \phi)^2}. \quad (1.134)$$

The second equality follows by differentiating $\sum_{k \geq 0} \phi^k = (1 - \phi)^{-1}$ and multiplying by ϕ . As ϕ approaches one, modest batches badly underrepresent the slow dependence. Lugsail changes the leading bias; it does not manufacture the missing long-time information.

Subtracting covariance estimates can produce a negative scalar estimate or negative eigenvalues in finite samples. Such an estimate cannot be inserted into a square root or treated as zero Monte Carlo uncertainty. Numerical validity, adequate batch counts, sensitivity to batch length and additional sampling remain necessary. A declared positive-definiteness correction is a different estimator and needs its own justification. The lugsail paper discusses these finite-sample issues; its asymptotic results are not guarantees of finite-run coverage.

BayesFilter already implements the lugsail option in `bayesfilter/inference/hmc_precision.py`, and posterior assessment uses the precision machinery. The remaining task is to establish that each consumer supplies appropriate retained draws, observables and batch settings. A separate experimental use of lugsail for acceptance uncertainty did not pass its strong-dependence promotion study. That result does not remove the posterior MCSE estimator, and the existence of the estimator does not qualify the experimental acceptance policy.

1.10.3 Modern $\hat{R}$ and many short chains

For m chains of n draws, define chain means $\bar{h}_j$, sample variances s_j^2 , grand mean $\bar{h}$, and

$$W = \frac{1}{m} \sum_j s_j^2, \quad B = \frac{n}{m-1} \sum_j (\bar{h}_j - \bar{h})^2, \quad \hat{V} = \frac{n-1}{n} W + \frac{B}{n}. \quad (1.135)$$

The classical statistic $\hat{R} = \sqrt{\hat{V}/W}$ compares between-chain disagreement with variation within chains. A common mean can hide differing scales or heavy tails, so modern practice first splits each chain, then applies the statistic to rank-normalized draws. For S pooled split-chain draws with midrank r_i , the normal score is

$$z_i = \Phi^{-1} \left(\frac{r_i - 3/8}{S + 1/4} \right). \quad (1.136)$$

Here Φ is the standard-normal CDF, so its inverse acts on probabilities strictly between zero and one. Apply the same calculation to the ranks of $|h_i - \text{median}(h)|$ and take the maximum of the rank and folded-rank results [Vehtari et al., 2021]. The folding checks scale disagreement that a location diagnostic can miss. Constant chains and undefined ratios require explicit treatment rather than automatic acceptance. Even agreement among dispersed chains cannot establish the absence of an unvisited mode.

When hundreds of chains each retain only a few draws, ordinary within-chain variance is poorly estimated. Nested $\hat{R}$ uses K superchains, each containing M subchains sharing an initial state X_{0k} but using independent randomness conditional on that state

[Margossian et al., 2024, Definition 2.2 and Theorem 3.1]. Let H_{km} be a subchain's mean after the chosen warm-up and $\bar{H}_k = M^{-1} \sum_m H_{km}$. If $\mu_n(x) = \mathbb{E}[H_{km} \mid X_{0k} = x]$ and $v_n(x) = \text{Var}(H_{km} \mid X_{0k} = x)$, the law of total variance yields

$$\text{Var}(\bar{H}_k) = \text{Var}\{\mathbb{E}[\bar{H}_k \mid X_{0k}]\} + \mathbb{E}\{\text{Var}(\bar{H}_k \mid X_{0k})\} \quad (1.137)$$

$$= \text{Var}\{\mu_n(X_{0k})\} + \frac{1}{M} \mathbb{E}\{v_n(X_{0k})\}. \quad (1.138)$$

Replication reduces the second term but leaves the first unchanged. This is why shared starts within a superchain are useful: independently dispersing every subchain would also average away the variation due to initialization.

For N retained draws h_{kmn} per subchain, the paper defines

$$\hat{B}_\nu = \frac{1}{K-1} \sum_k (\bar{H}_k - \bar{H})^2, \quad (1.139)$$

$$\tilde{B}_k = \frac{1}{M-1} \sum_m (H_{km} - \bar{H}_k)^2, \quad \tilde{W}_k = \frac{1}{M} \sum_m \frac{1}{N-1} \sum_n (h_{kmn} - H_{km})^2, \quad (1.140)$$

$$\widehat{W}_\nu = \frac{1}{K} \sum_k (\tilde{B}_k + \tilde{W}_k), \quad \hat{R}_\nu = \sqrt{1 + \hat{B}_\nu / \widehat{W}_\nu}. \quad (1.141)$$

Set $\tilde{B}_k = 0$ for $M = 1$ and $\tilde{W}_k = 0$ for $N = 1$; at least one of M, N must exceed one, and $K > 1$. The behavior and cutoff depend on the design, particularly M and N . A familiar long-chain cutoff cannot simply be copied to an arbitrary ensemble. If subchains share adaptive statistics, conditional independence need not hold and the second line of (1.138) acquires cross-covariance terms. An independently learned, frozen kernel followed by a correctly designed diagnostic run is a simpler first application.

The first term in (1.138) measures variation of the conditional expectation across starts, not its distance from μ . A constant but wrong $\mu_n(x)$ has zero such variation. Nested $\hat{R}$ therefore remains a diagnostic whose usefulness depends on informative initial dispersion and the chain dynamics, not a proof of zero bias.

1.10.4 A stopping rule must name the economic quantity

For a scalar mean with a valid central limit approximation, one can require an estimated confidence half-width

$$z_{1-\alpha/2} \sqrt{\hat{\sigma}_h^2/n} \leq \delta_{\text{abs}} + \delta_{\text{rel}} s_h, \quad (1.142)$$

where the tolerances are chosen for the scientific use and s_h is a specified scale, such as the posterior standard deviation. A relative tolerance based only on $|\hat{\mu}|$ is unstable near a zero effect. Fixed-width stopping has an asymptotic theory under explicit assumptions; arbitrary repeated inspection of nominal fixed-sample intervals does not give a finite-sample sequential guarantee [Flegal and Gong, 2013]. Readiness checks, minimum information, a finite resource limit and an explicit inconclusive outcome are part of a usable procedure.

For a smooth nonlinear function $g(\mu)$, a delta-method variance is $\nabla g(\mu)^\top \Sigma \nabla g(\mu)/n$. This describes $g(\bar{h})$; the posterior expectation $\mathbb{E}_\pi g(h(X))$ is instead estimated by averaging $g(h(X_t))$ and needs that observable's long-run variance. For a quantile q_p with continuous positive marginal density $f_h(q_p)$, linearizing the empirical CDF gives

$$\hat{q}_p - q_p \simeq -\frac{\hat{F}_n(q_p) - p}{f_h(q_p)}, \quad \text{Var}(\hat{q}_p) \simeq \frac{\sigma_{\mathbf{1}\{h \leq q_p\}}^2}{n f_h(q_p)^2}. \quad (1.143)$$

Thus a mean MCSE is not a quantile MCSE. For an event probability use the indicator’s dependence, not an independent-binomial standard error unless independence is justified. Zero observed rare events produce an empirical variance of zero while leaving the probability uncertain. More sampling, an independently justified rare-event method, or an explicitly unresolved probability is needed. Heavy-tailed targets can lack the moments required for variance-based criteria; bounded observables and suitably justified quantile procedures may then be more appropriate.

These distinctions preserve the intended tuning interface. Acceptance compatibility, finite numerical behavior and movement determine which fixed-kernel candidates remain usable for further assessment. $\hat{R}$ must not be reinstated as an acceptance-tuning requirement. Warm-up and posterior assessment can use rank diagnostics, precision and reference agreement in their own roles. Choosing an epsilon and a trajectory length is not a claim that the retained posterior has converged.

1.11 Practical choices for structural models

The least expensive useful change depends on the measured bottleneck. A representative MacroFinance MIDAS adapter differentiates a batched, masked Kalman likelihood with correlated noises and seven sampled parameters. There the observation horizon can dominate the work even though the parameter dimension is small. A stochastic-volatility benchmark with thousands of sampled latent states does not measure the same problem. The inspected BGS description in `dsge_hmc` instead declares an endpoint potential based on a particular UKF approximation and a deterministic position-only proposal field that is not asserted to be its exact score. That distinction affects the admissible methods before speed is considered.

A reversible shear integrator with an exact endpoint MH correction can remain valid with a position-only non-gradient force, as derived earlier. The equipartition identity and the stationary microcanonical dynamics use the actual score and cannot inherit that conclusion. BGS therefore needs either an eligible total score of its declared endpoint target or a separate derivation and validation of the proposed force-based method. Relabeling the field as a gradient would be wrong. The current capability registry describes that branch as conditional mechanics without exact-score retained-member authority. The operational reference and registry remain the entry points for an HMC consumer; this chapter does not add another tuner.

Table 1.2: Method selection depends on the sampled target and the measured cost. These are proposed investigations, not local speed rankings.

Intervention	Most relevant circumstance	First discriminating evidence
Validated whitening and noncentering	Strong correlation, a funnel, or varying conditional scale	Exact transformed density and total score; inverse round trips; model-coordinate tails after freezing the map.

Intervention	Most relevant circumstance	First discriminating evidence
Randomized HMC and SNAPER	Resonance or costly trajectory selection on batched hardware	Same target, map and starts as corrected HMC; slow projections and squared observables; total time to declared precision.
Equivalent target/score optimization	Kalman, UKF or model-solution work dominates a gradient	Value and total-gradient parity, including masks, noise dependence and normalization; measured compilation and batch latency.
MAMS	Integration work remains costly after geometry repair	Sphere-measure correction, stable force flow, reversibility, reference agreement and equal-accuracy total gradient/time comparison.
LAPS	Large ensembles exhibit favorable measured batch scaling	Mean-shift and covariance counterexamples; independent warm-up assessment; total work and latency including its transient.
MALT or MEADS	Persistent dynamics may reduce residual resonance or random-walk behavior	Refreshment and momentum/slice-state correctness; cost of rejections and ensemble coordination.
Pathfinder	Initial exploration or geometry learning dominates	Initialization quality across dispersed starts, modes and boundaries; subsequent exact-kernel behavior and total setup cost.
Delayed rejection	Residual local stiffness survives a useful map	Reverse rejection factors and retry accounting; difficult-region movement versus extra gradients.
Particle-MALA/mGRAD	The latent trajectory itself must be sampled	Conditional smoothing reference, correct joint auxiliary weights, path mixing and scaling in horizon/state dimension.

For the present TensorFlow setting, randomized-length HMC followed by SNAPER is the first sampler comparison worth making. The installed TFP source already supplies the kernel and trajectory adaptation, although that does not integrate them with BayesFilter’s evidence and frozen-kernel interfaces. Whitening should continue alongside this work. Where a marginal Gaussian likelihood is expensive, investigate an equivalent parallel formulation or target/score optimization before attributing the cost to the sampler. MAMS is the next major kernel candidate. LAPS becomes attractive if those same model operations scale well with a large chain batch and its transient has independent diagnostic support.

The proposed order reflects implementation availability and mathematical applicability, not measured superiority. It can change after profiling. Custom QZ decompositions, equilibrium-solution derivatives, boundary handling and filter implementations make a

JAX migration a separate engineering project. An author implementation in JAX is valuable for independent checks and algorithm details; it does not make a TensorFlow port either trivial or impossible.

1.12 Discriminating tests and model comparisons

Two comparisons are needed. A comparison with *matched geometry* fixes the posterior, data, numerical precision, coordinate map and model-coordinate starts, then allows each method its own valid tuning under a comparable resource budget. It measures the transition method. An *end-to-end* comparison also includes compilation, initialization, map fitting, tuning, fresh verification and retained sampling. It measures what a user pays. Both should include the current corrected broad-grid HMC, a simple randomized-length version, and correctly configured NUTS before crediting a more elaborate method. Count every force evaluation, including integrator substages and rejected delayed-rejection proposals.

The primary question is time to a predeclared accuracy for stated posterior quantities. Numerical validity, reference disagreement, divergences and missing diagnostics can reject a candidate. Acceptance, ChEES/SNAPER objectives, equipartition and short-run ESS explain behavior but do not replace posterior checks. Paired repeated runs and uncertainty for the declared comparison are necessary for a supported ranking; passing a hard screen leaves a method viable, not superior.

Table 1.3: A model ladder that isolates mechanisms before expensive state-space applications.

Model or construction	Misleading success	Discriminating observation
Gaussian at a full or half Hamiltonian period	Acceptance one; apparent movement from sign changes	Means, variances and squared-observable correlation; exact independent Gaussian reference and random-length control.
Rotated anisotropic Gaussian	Good average movement while one direction remains slow	Known covariance, eigenvector projections, worst declared estimand and diagonal versus full whitening.
Shifted Gaussian with correct covariance	Centered equipartition equals its target value	Known mean error, full diagnostic and separately identified implementation variant.
Wrong-correlation Gaussian or equal-variance mixture	Correct marginal variances hide dependence or distribution shape	Cross moments, fourth moments, event probabilities and full-matrix checks.
Exactly and partially whitened funnel	Good latent diagnostics hide map errors or model-coordinate tails	Analytic map and Jacobian; total score; model-coordinate moment and tail reference; residual curvature.

Model or construction	Misleading success	Discriminating observation
Heavy-tailed scale model	An apparently stable variance estimate assumes a nonexistent moment	Explicit moment conditions; bounded functions and quantiles; insufficient information as a valid outcome.
Separated mixture with unbalanced starts	All chains agree inside the same mode	Known mode weights, cross-mode transitions and independently dispersed initializations.
Constrained parameter near a boundary	Finite target values conceal a missing Jacobian or saturated score	Analytic or quadrature reference and directional derivatives near the boundary; appropriate discontinuity treatment.
LGSSM with masks and correlated noise	A faster filter evaluates a different likelihood	Sequential/parallel value and total-score agreement before posterior comparison; short/long horizons.
Nonlinear stochastic volatility or other nonlinear SSM	Samplers agree because they share the same biased filter	Small independent reference; separate likelihood-approximation error; conditional latent-path tests where applicable.
Representative MIDAS and BGS scopes	Synthetic performance does not transfer to model-solution work	Actual target/force identity, measured batch scaling, boundary and weak identification regimes, economic estimands.

Transition-level tests should precede this ladder. For MAMS they include the excluded one-dimensional formula, zero force, velocities aligned and antialigned with the force, preservation of unit norm, forward/reverse round trips, analytic versus finite-difference sphere Jacobians, and large-argument stability. For MALT, compare the sum of local integration errors with the full forward/reverse path-density ratio. For delayed rejection, test a case in which the reverse first-stage rejection probability differs sharply from the forward one. For Gaussian message passing, dropping one log-normalization term must fail a likelihood test even if the filtered mean still matches.

Posterior checks can use exact Gaussian sampling, numerical quadrature on small parameter targets and independent state-space reference calculations. Simulation-based calibration complements these checks by repeatedly drawing parameters and data from the model and assessing posterior ranks [Talts et al., 2020]. If the data-generating parameter and the posterior draws are jointly exchangeable conditional on the data, the parameter’s rank is uniform with randomized tie-breaking; correlated MCMC draws require the corresponding dependence treatment before a rank test is interpreted. Calibration averaged over the prior can also conceal a failure in a rare but important regime, so boundary, weak-identification and long-horizon cases need explicit coverage. MCMC implementation tests based on invariance or reversible simulation provide another layer

[Gandy and Scott, 2021]; no one finite suite proves correctness for every model.

This design makes failures informative. A failed sphere Jacobian is an implementation error. Correct transitions with poor model-coordinate tails after partial whitening indicate residual geometry or insufficient sampling. A correct sampler for the declared UKF posterior can still disagree with a more accurate likelihood because the target approximation differs. A method that passes all correctness checks but uses more total work is viable and possibly useful for latency, although a claimed work reduction would be unsupported. These distinctions determine the next repair.

1.13 Sources, versions and the scope of the conclusions

The literature review behind this chapter inspected the technical methods, relevant theory, experiments and appendices of the cited recent papers, as well as author implementations where available. The two motivating papers are the May 2025 revision of MAMS, arXiv:2503.01707v2, and the January 2026 LAPS preprint, arXiv:2601.16696v1. The unadjusted-bias discussion uses the June 2026 revision of arXiv:2412.08876; its original identifier year alone does not identify the examined argument. Bibliography entries record the inspected versions when a working paper is used.

Local copies, extracted text and source hashes are preserved under `.localresources/hmc-methods-review-20261007/`.

The source manifest records pinned revisions of BlackJAX and the authors' sampler benchmarks. It also records the TFP source, installed TFP 0.25.0 files, Stan Pathfinder, Particle-MALA, delayed-rejection GHMC and nested- $\hat{R}$ sources. The implementation differences described in the LAPS and SNAPER sections refer to those snapshots. They must not be read as assertions that all versions implement the same equations.

The algebra in this chapter identifies preserved measures, required corrections and diagnostic limitations. Symbolic checks accompany the document review, but cannot establish geometric ergodicity, finite-time convergence, differentiability of an economic solution operator, or practical speed on a new consumer. The resulting recommendation is specific: first combine validated geometry with inexpensive randomized trajectory adaptation and an equivalent efficient likelihood; then evaluate MAMS, and evaluate LAPS when measured ensemble scaling warrants it. The state-space tests above are what would turn that recommendation into evidence for these repositories.

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